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ASTRO 320W

Astro 320W Midterm Observing Report

Introduction

In this report, I present an analysis of CCD calibration and stellar imaging observations obtained by the Artemis group with the Davey Laboratory 24-inch telescope, following the course manual [1]. The observations include bias, dark, and flat-field frames, together with stellar focus sequences, repeated photometric exposures, BVR imaging, and double-star fields. I use these data to characterize the detector, examine the stability of the photometric measurements, and determine the camera's plate scale. The observations were collected with the group; I carried out the analysis and wrote the report.

1 Exercise A: CCD Characteristics

Digital number (DN) and analog-to-digital unit (ADU) refer to the same signal unit. Standard deviation (SD) describes measurement scatter; standard error of the mean (SEM) describes the estimated uncertainty in a mean. Unless noted otherwise, regional measurements use a central 100×100-pixel square starting at zero-based array coordinates $(1478, 1478)$ in the unbinned 3056×3056 images.

1.1 Bias Frames and Read Noise

We collected 20 bias frames (0.1-s dark exposures) at the coldest cooler setting. Two instructor-provided Python routines calculate

$$\sigma_t(x, y) = \sqrt{\frac{1}{N}\sum_{i=1}^{N}[B_i(x, y) - \overline{B}(x, y)]^2}, \quad (1)$$

then average over the detector or a selected region. This measures temporal variation rather than fixed pixel-to-pixel bias structure. The full-detector result was 9.645 DN with $N = 20$ in the denominator (9.896 DN with $N - 1$); the central result was 9.655 DN.

To examine the contribution from fixed bias structure, I formed a 32-bit average master bias in AstroImageJ and subtracted it from each frame. The mean spatial SD decreased from 13.847 to 9.790 DN (Table 1), indicating that a substantial fraction of the single-frame scatter is associated with a stable spatial pattern. Because each frame also contributes to the master, the residual variance is reduced:

$$\mathrm{Var}(B_i - \overline{B}) = \sigma_{\mathrm{RN}}^2(1 - 1/N). \quad (2)$$

Correcting for this gives $9.790/\sqrt{19/20} = 10.04$ DN. I independently estimated the read noise by subtracting ten disjoint pairs of raw bias frames. Subtraction removes the fixed spatial pattern, while the independent noise variances of the two frames add:

$$R_{\mathrm{DN}} = \sigma_{B_1 - B_2}/\sqrt{2}. \quad (3)$$

The paired read-noise results were

$$\boxed{R_{\mathrm{full}} = 10.045 \pm 0.007 \text{ DN}}, \quad (4)$$

$$\boxed{R_{\mathrm{central}} = 10.045 \pm 0.019 \text{ DN}}. \quad (5)$$

These are SEMs of ten pair estimates; sample SDs were 0.021 and 0.060 DN, respectively. Table 2 lists the pair measurements. The three unassigned central SDs were 14.286, 14.324, and 14.246 DN; all enter the mean. The predicted shared-master residual, 9.791 DN, agrees with 9.790 DN. I adopt the paired full-detector result because it does not depend on a shared master. To investigate whether these properties vary across the detector, I divided the images into an 8×8 grid and repeated the paired measurement within each block (Figure 1). Block read noise ranges from 9.63 to 10.48 DN. In contrast, the

Table 1: Full-image spatial standard deviations from 20 bias frames. Sample SD describes scatter among the measured standard deviations. Nominal SEM is sample SD divided by $\sqrt{20}$. The corrected measurements share a master bias and are not independent.

Images	Mean spatial SD (DN)	Sample SD (DN)	Nominal SEM (DN)
Raw bias frames	13.847	0.092	0.021
Bias-subtracted frames	9.790	0.027	0.006

Table 2: Raw-bias pair differences and inferred read noise. Central measurements use the same 100×100 pixels. All values are in DN. The last three central entries are not assigned to specific pairs because their export does not preserve pair labels.

Pair	Full σ_{diff}	Central σ_{diff}	Full R	Central R
1–2	14.211	14.187	10.049	10.031
3–4	14.199	14.058	10.040	9.940
5–6	14.267	14.243	10.088	10.071
7–8	14.173	14.100	10.022	9.970
9–10	14.209	14.209	10.047	10.047
11–12	14.193	14.263	10.036	10.086
13–14	14.173	14.141	10.022	9.999
15–16	14.245	—	10.073	—
17–18	14.190	—	10.034	—
19–20	14.197	—	10.039	—

mean single-frame spatial SD ranges from 9.67 to 61.85 DN, with a strong localized excess. Stable bias structure and pixels with unusually large offsets contribute to the spatial scatter, whereas their fixed contribution cancels in the pair differences. The contrast between the two maps therefore indicates that single-frame bias scatter cannot be attributed to read noise alone.

The measured Exercise B gain applies to 2×2 binning. For these 1×1 frames I use the header gain, $g_{\mathrm{hdr}} = 1.34\,\mathrm{e^-\,DN^{-1}}$. The temporal, shared-master, and paired results correspond to 12.9, 13.1, and $\boxed{13.5\ \mathrm{e^-\ pixel^{-1}\ read^{-1}}}$, respectively. The paired SEM alone contributes 0.009 electron; the header gain's unknown uncertainty limits the electron conversion.

1.2 Dark Signal and Exposure Time

I measured six exposures from 0.1 to 60 s at a reported 0°C setpoint. Their actual CCD temperatures were not recorded. The headers contain neither `CALSTAT` nor `PEDESTAL`, so I treated them as uncorrected. Using the central master-bias mean $B = 1038.930$ DN, I calculated

$$Q = g_{\mathrm{hdr}}(D_{\mathrm{raw}} - B), \qquad I = Q/t. \tag{6}$$

The master bias was obtained at a −30°C setpoint, so a temperature-dependent offset may remain after subtraction. The telescope was also uncovered, allowing possible stray-light contamination. Consequently, the measured signal may contain contributions in addition to intrinsic dark current.

For a formal uncertainty in the regional mean, I used independent pixel shot noise, read noise, and the finite 20-frame master:

$$\sigma_Q = \frac{g_{\mathrm{hdr}}}{\sqrt{n}} \sqrt{\frac{\max(D_{\mathrm{net}}, 0)}{g_{\mathrm{hdr}}} + R^2(1 + 1/N)}. \tag{7}$$

Here $n = 10000$, $N = 20$, $R = 10.045$ DN, and $\sigma_I = \sigma_Q/t$. This model excludes hot-pixel variability, spatial correlations, bias-temperature mismatch, gain uncertainty, and stray light. The

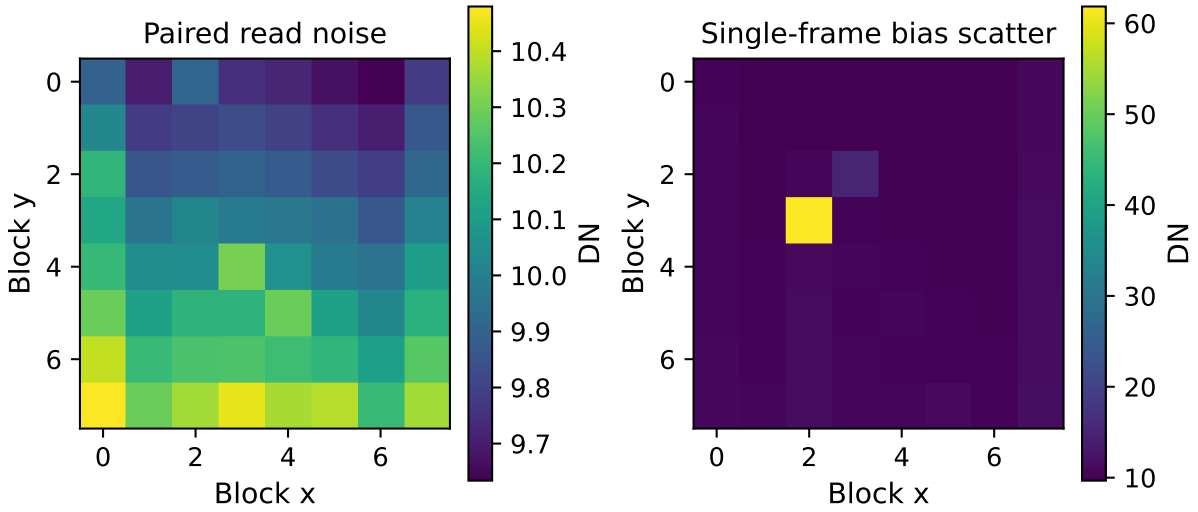


Figure 1: Read noise from ten disjoint bias pairs (left) and mean single-frame spatial SD (right), measured in the same 8×8 grid. Array block indices increase to the right and downward. The panels use separate DN scales.

shared master also correlates different frames. Table 3 lists the formal errors; the systematic uncertainties remain larger.

The 0.1-s frame gives an apparent rate of 423 $\mathrm{e^-\ pixel^{-1}\ s^{-1}}$. A small offset divided by a short exposure can produce this result, so I show it but exclude it from the fit. For 6–60 s, the mean rate is

$$\boxed{\bar{I} = 13.76 \pm 0.51\ \mathrm{e^-\ pixel^{-1}\ s^{-1}}}. \tag{8}$$

The uncertainty is the SEM; the sample SD is 1.14. An unweighted fit $Q = mt + b$ gives

$$m = 14.33 \pm 0.78\ \mathrm{e^-\ pixel^{-1}\ s^{-1}}, \tag{9}$$

$$b = -14.8 \pm 28.4\ \mathrm{e^-\ pixel^{-1}}. \tag{10}$$

The quoted fit errors are estimated from the residual scatter. The approximately linear increase in charge is consistent with a nearly constant accumulation rate over 6–60 s. However, stray light adds signal, while the unmatched bias can shift the result in either direction. The fitted rate therefore describes the observed signal accumulation rather than a strict upper limit on intrinsic dark current.

1.3 Temperature Dependence

Four frames have verified 300-s exposures and recorded starting temperatures: −7.84, 0.84, 18.75, and 20.22°C (Table 4). The files labeled `-20C.fit` and `-25C.fit` instead contain 0.12-s and 10-s exposures without recorded CCD temperatures. I excluded them and used the same region, bias, and gain for the four retained frames.

The measured rate increases from 0.042 to 3.946 $\mathrm{e^-\ pixel^{-1}\ s^{-1}}$ across the retained temperature range. However, the two warmest frames differ by almost a factor of three in signal despite starting only 1.47°C apart. This suggests that the starting temperature may not represent the conditions throughout each exposure. The coldest measurement is also particularly sensitive to the bias subtraction. To examine whether the overall trend is consistent with thermal generation, I fitted

$$I(T) = I_0 \exp[-A/(kT)], \tag{11}$$

$$\ln(I/I_{\mathrm{ref}}) = c - (A/k)(1/T), \tag{12}$$

where T is in kelvin, $k = 8.617333 \times 10^{-5}\ \mathrm{eV\,K^{-1}}$, and $I_{\mathrm{ref}} = 1\ \mathrm{e^-\ pixel^{-1}\ s^{-1}}$. The unweighted slope is approximately -1.107×10^4 K, giving $\boxed{A_{\mathrm{app}} = 0.95 \pm 0.14\ \mathrm{eV}}$. The er-

Table 3: Central-region dark signals using the nominal header gain. Errors describe only the independent-pixel model; systematic errors are excluded.

t (s)	Net signal (DN)	Q (e$^-$ pixel^{-1})	I (e$^-$ pixel^{-1} s^{-1})
0.1	31.568	42.30 ± 0.15	423.0 ± 1.5
6	58.079	77.83 ± 0.16	12.971 ± 0.027
10	113.585	152.20 ± 0.19	15.220 ± 0.019
30	279.963	375.15 ± 0.24	12.5050 ± 0.0079
45	452.549	606.42 ± 0.28	13.4759 ± 0.0063
60	654.913	877.58 ± 0.33	14.6264 ± 0.0054

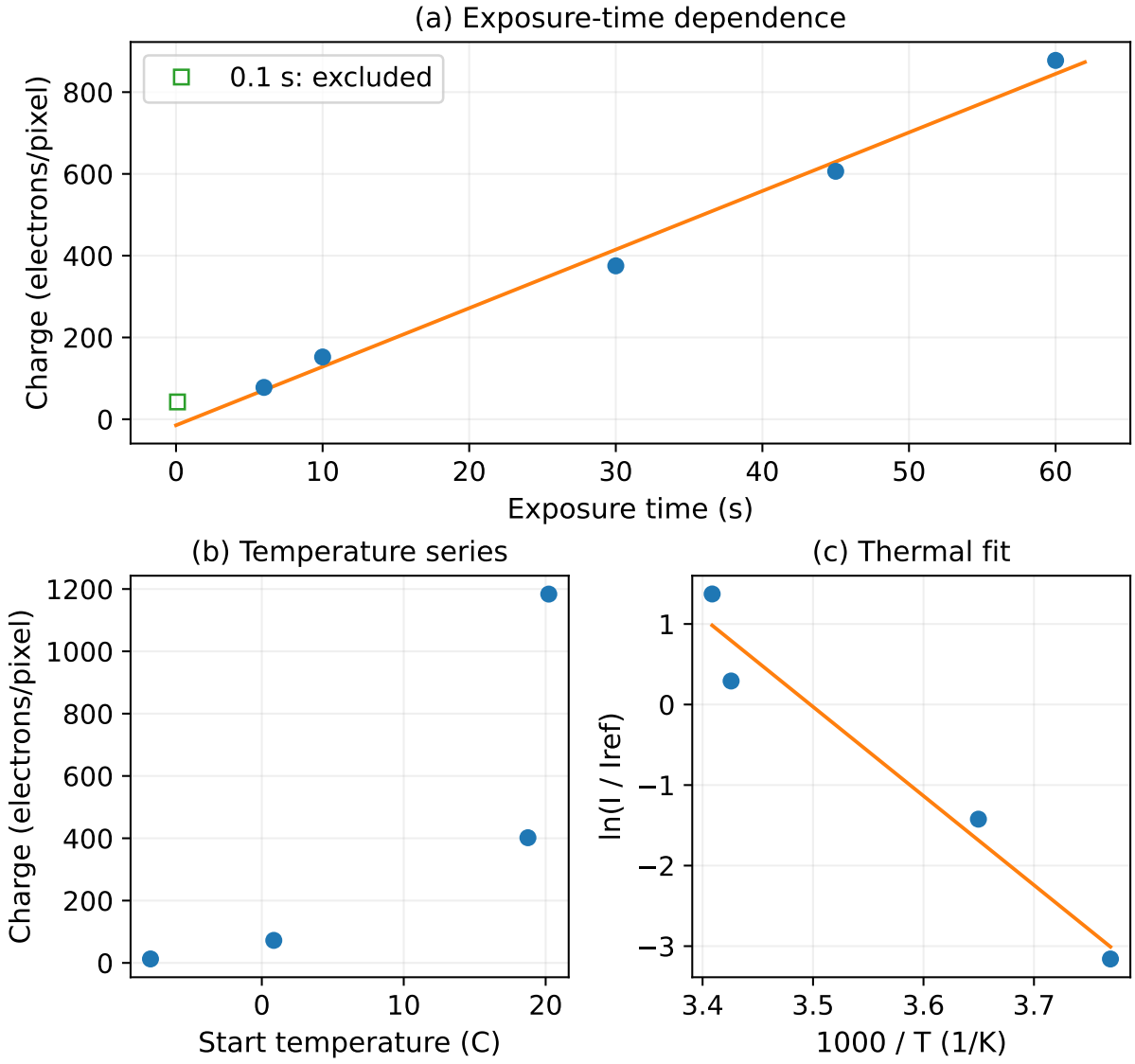


Figure 2: Dark-signal measurements: (a) exposure-time dependence, with the 0.1-s point excluded from the fit; (b) charge versus starting temperature for four 300-s frames; (c) the linearized thermal relation. Formal regional-mean errors are smaller than the markers and exclude calibration and temperature systematics.

ror follows from the regression slope error, $\sigma_A = k\sigma_{\mathrm{slope}}$. The increase is consistent with thermal generation, but the small sample, unmatched bias, and uncertain temperatures limit the activation-energy estimate.

2 Exercise B: Flat Fields and Gain

2.1 Photon Transfer Curves

I measured regional means and spatial SDs in AstroImageJ. The sky sequence contains twenty 6-s R-band frames with regional means of 16048–38285 DN. All measured regions were digitally unclipped; one frame has three 65535-DN pixels elsewhere, leaving nineteen entire images free of digital clipping. The dome sequence contains 13 frames at seven exposures (6, 12, 15, 18, 30, 45, and 90 s), with changing lamp settings. Its headers record acquisition-time dark subtraction and `PEDESTAL=-100`. The original PTC region and pedestal handling were not saved.

For photon-dominated signal S, the variance follows $\sigma^2 \simeq S/g$, giving an expected log–log noise-versus-signal slope of $1/2$. The sky measurements rise steadily with signal, whereas the dome measurements turn over at high signal (Figure 3). Because the measured spatial SD also contains illumination gradients and pixel-sensitivity structure, these curves provide a qualitative comparison rather than an independent precision gain measurement. The dome points were digitized from the preserved plot and are not used in a numerical fit.

2.2 Gain from Paired Flats

We took twenty 0.2-s and twenty 10-s dome flats with 2×2 binning. They were dark-corrected during acquisition. I subtracted the stored 100-DN pedestal once in 32-bit arithmetic, then measured ten disjoint pairs at each exposure in a central 100×100 region starting at $(714, 714)$, without alignment or normalization.

If photon noise dominates, the gain estimate is

$$g_{\mathrm{ph}} = \frac{\overline{F}_1 + \overline{F}_2}{\sigma^2_{F_1 - F_2}}. \tag{13}$$

For the first 10-s pair, means of 27292.439 and 27332.640 DN and a difference SD of 206.305 DN give $1.283\ \mathrm{e^-\ DN^{-1}}$. Table 5 lists all pairs. The 10-s mean is $1.275 \pm 0.006\ \mathrm{e^-\ DN^{-1}}$ (SEM; SD 0.019); the 0.2-s mean is $0.829 \pm 0.005\ \mathrm{e^-\ DN^{-1}}$ (SD 0.014). The lower value from the faint flats is expected when additive noise contributes a substantial fraction of the measured variance. This effect biases the photon-only estimate downward without requiring a change in the physical gain.

To account for this additive contribution, I fitted all twenty pairs with a linear variance model:

$$S = (\overline{F}_1 + \overline{F}_2)/2, \tag{14}$$

$$V = \sigma^2_{F_1 - F_2}/2 = aS + c, \qquad g = 1/a. \tag{15}$$

The unweighted fit gives $a = 0.7751$, $c = 251\ \mathrm{DN^2}$, and

$$\boxed{g_{2\times2} = 1.290 \pm 0.006\ \mathrm{e^-\ DN^{-1}}}. \tag{16}$$

The error is σ_a/a^2; it excludes calibration and region-selection systematics. The intercept includes read and calibration noise. Intermediate signal levels would provide a stronger linearity test.

2.3 Flat-Field Structure

I compared a 6-s sky flat, `AutosaveImage-0002skyflat.fit`, with the 10-s dome flat `DOME_FLAT_20_EXP-000110.fit`. For the dome image I removed the stored pedestal once. Python measured horizontal and vertical profiles as the median across central 11-pixel-wide strips, then normalized by each frame's median (Figure 4). The full horizontal profiles span 0.41–1.05 for sky and 0.34–1.07 for dome; vertical profiles span 0.16–1.05 and 0.11–1.07. Both show reduced signal near the edges and structure across the field. Normalization allows comparison of profile shapes despite the different overall

Table 4: Four verified 300-s temperature exposures. Errors use the same formal regional-mean model as the time series and exclude temperature and calibration systematics.

T_{start} (°C)	Net signal (DN)	Q (e^- pixel^{-1})	I (e^- pixel^{-1} s^{-1})
-7.84	9.504	12.74 ± 0.14	0.0425 ± 0.0005
0.84	53.934	72.27 ± 0.16	0.2409 ± 0.0005
18.75	299.826	401.77 ± 0.24	1.3392 ± 0.0008
20.22	883.360	1183.70 ± 0.37	3.9457 ± 0.0012

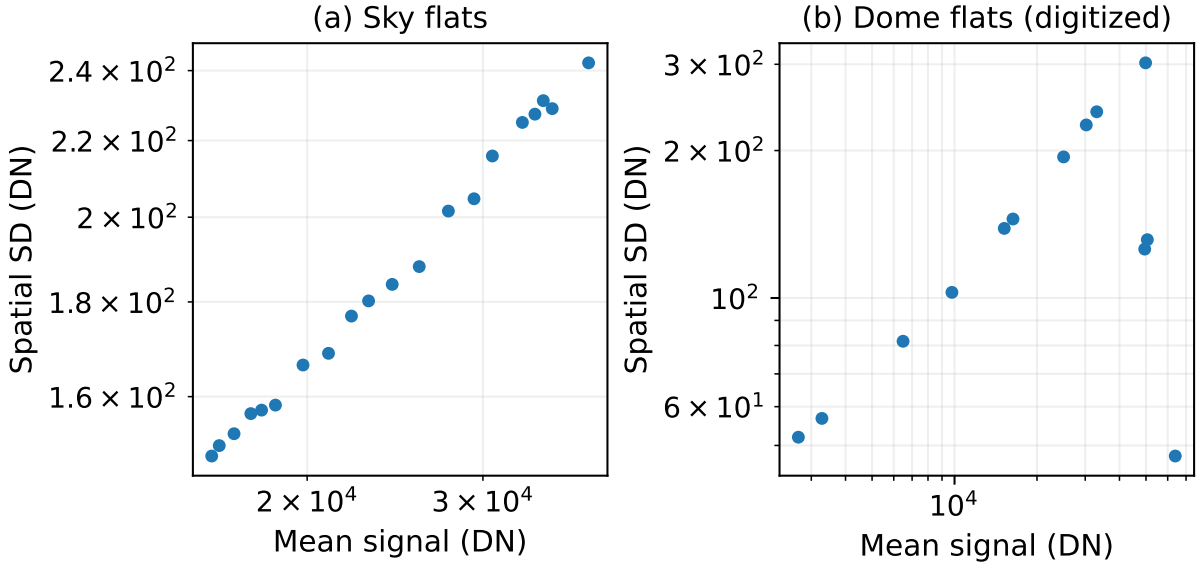


Figure 3: Sky and dome photon transfer curves. Both axes are logarithmic; noise is regional spatial SD. Sky points are original numerical measurements; dome points were digitized from the preserved plot. No regression uses the digitized values.

Table 5: Photon-only paired-flat gain estimates, in e^- DN^{-1}. This estimator does not remove read noise.

Pair	0.2-s flats	10-s flats
1–2	0.836	1.283
3–4	0.804	1.274
5–6	0.836	1.264
7–8	0.854	1.232
9–10	0.810	1.280
11–12	0.834	1.271
13–14	0.825	1.294
15–16	0.831	1.289
17–18	0.828	1.296
19–20	0.837	1.270

brightnesses.

Sky flats follow the science light path but require twilight and can contain stars and changing sky gradients. Dome flats can be taken after dark under controlled illumination, although the screen may illuminate the optics differently. Both show gradients and optical structure (Figure 5).

2.4 Saturation Limits

Saturation can occur in the pixel well, analog electronics, or ADC. The 16-bit digital ceiling is 65535 DN. At a dome-flat mean of 64376 DN, spatial SD falls to 63 DN (Table 6), consistent with clipping or nonlinearity. Its maximum of 64457 DN is close to the ceiling, but the original offset handling is unknown. The STX-

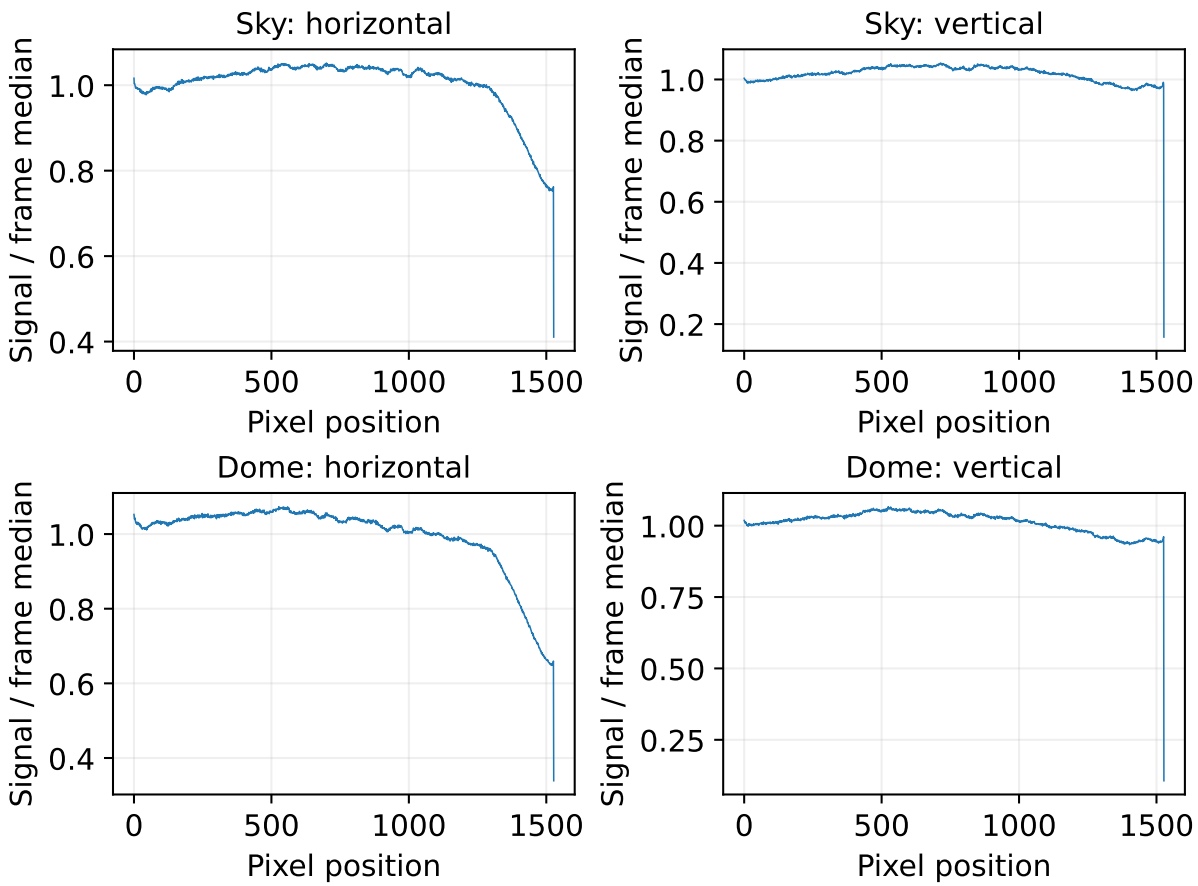


Figure 4: Horizontal and vertical profiles for explicitly identified sky (top) and dome (bottom) frames. Each is the median across a central 11-pixel-wide strip, divided by the frame median.

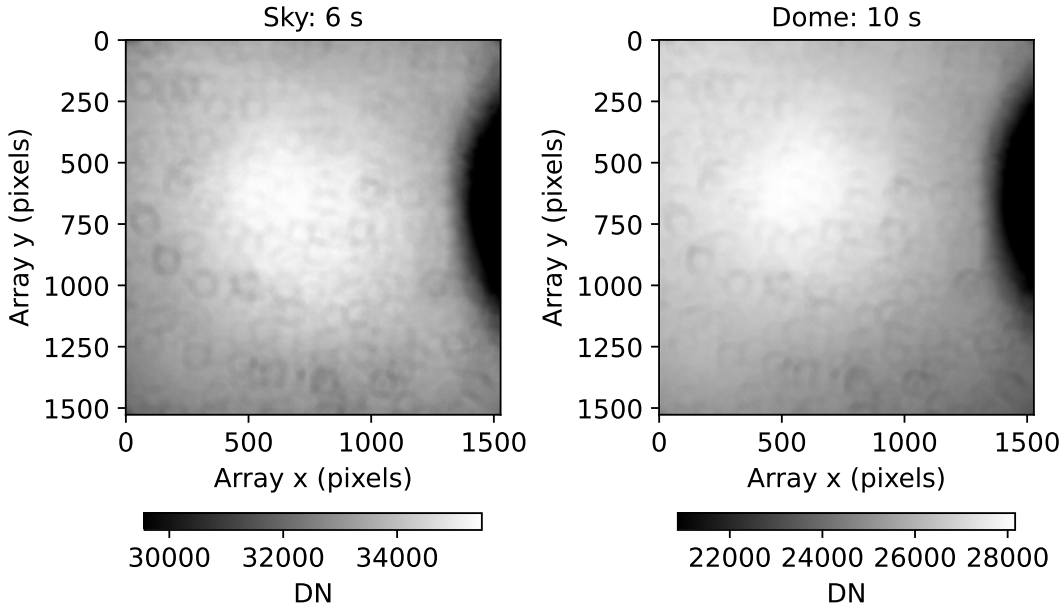


Figure 5: The same identified sky and dome frames used for the profiles, with separate DN scales. Ring shadows and edge losses are visible in both frames.

9000 specifications list a KAF-09000 sensor with nominal 110000 e^- full well and a 16-bit ADC [2]. At the measured binned gain, $65535g \simeq 8.45 \times 10^4$ e^-. The ADC is therefore the expected limit before the nominal pixel well fills, provided the analog chain stays linear. An earlier analog limit remains possible because I did not measure the electronics or full-well capacity independently.

3 Exercise C: Stellar Images and Photometry

3.1 Observations and Focus

We observed Omicron Aquilae for focus and Ogma (HD 149026) for repeatability and exposure-time tests through Johnson R. Wind introduced pointing disturbances of up to about 5 arcseconds. AstroImageJ calibration subtracted a master dark containing bias, then divided by a normalized R-band dome flat:

$$I_{\mathrm{cal}} = (I_{\mathrm{raw}} - D_{\mathrm{master}})/F_{\mathrm{norm}}. \tag{17}$$

I did not subtract a separate bias. Focus and repeatability use 2×2 binning; the exposure-time series is unbinned.

The focus sequence used 6-s exposures, stepping from 17000 toward 15000 μm. Table 7 lists the AstroImageJ FWHMs; Image 5 is absent from the saved sequence. FWHM decreases and then increases, locating best focus near $\boxed{f_{\mathrm{best}} \approx 16000\ \mu\mathrm{m}}$. Repeated-position means and sample SDs are 10.22 ± 0.66 pixels at 16000 and 11.72 ± 0.63 pixels at 17000. The scatter at repeated positions indicates that seeing, tracking, and profile measurement also affect the measured width, limiting the precision with which the focus minimum can be determined.

The raw focus frames have peaks of 49146–49575 DN and 4–94 pixels above 48000 DN in peak-centered 41×41 boxes. The flattened profiles are consistent with a nonlinear response (Figure 6). The sequence therefore identifies a best-focus region, but the roughly 10-pixel minimum should not be interpreted as the width of an unsaturated stellar image.

3.2 Photometric Repeatability

I measured fifteen individual 8-s Ogma frames with a 17-pixel aperture and 34–67-pixel sky annulus. For net counts C_i,

$$R_i = C_i/t_i, \tag{18}$$

$$D_i = 100(R_i - \bar{R})/\bar{R}, \tag{19}$$

$$s_R = \sqrt{\sum_i (R_i - \bar{R})^2/(N-1)}. \tag{20}$$

Figure 7 shows the count-rate deviations. The mean rate is 199592 DN s^{-1} and the sample SD is 13691 DN s^{-1}, giving $\boxed{100 s_R/\bar{R} = 6.86\%}$ scatter. The mean formal AstroImageJ error, 0.107%, is about 64 times smaller. However, Ogma's flattened 48919–49598-DN cores (Figure 8) make this an unsuitable test of unsaturated precision. The 6.86% is the scatter among individual frames, not the error on the mean.

To obtain a repeatability measurement that avoids the nonlinear Ogma cores, I measured an isolated, unsaturated field star in the same 15 calibrated frames, without stacking or alignment. Python iterated a positive-weight centroid in a 19×19 window, using the same aperture and annulus with median sky subtraction. Its first array position was (221.07, 1237.25) (Figure 9) and its peaks were 6332–9826 DN. Its mean rate was 10887 DN s^{-1}, with sample SD 145 and relative scatter $\boxed{100 s_R/\bar{R} = 1.33\%}$ (deviations −2.5 to +3.2%).

For an approximate statistical comparison I used

$$\sigma_C^2 \simeq C/g + N_a s_b^2 + \pi N_a^2 s_b^2/(2N_b), \tag{21}$$

with $g = 1.290$ e^- DN^{-1}, aperture and annulus pixel counts N_a, N_b, and $s_b = 1.4826\,\mathrm{median}|B - \mathrm{median}(B)|$. The final term approximates median-sky uncertainty for Gaussian noise. The mean modeled fractional error is $\boxed{0.89\%}$; source photon noise alone gives

Table 6: Original high-signal dome-flat measurements used to investigate saturation.

Mean (DN)	StdDev (DN)	Min (DN)	Max (DN)
49683.201	138.673	41776	49785
51474.074	152.154	46572	53144
64376.194	63.203	60650	64457

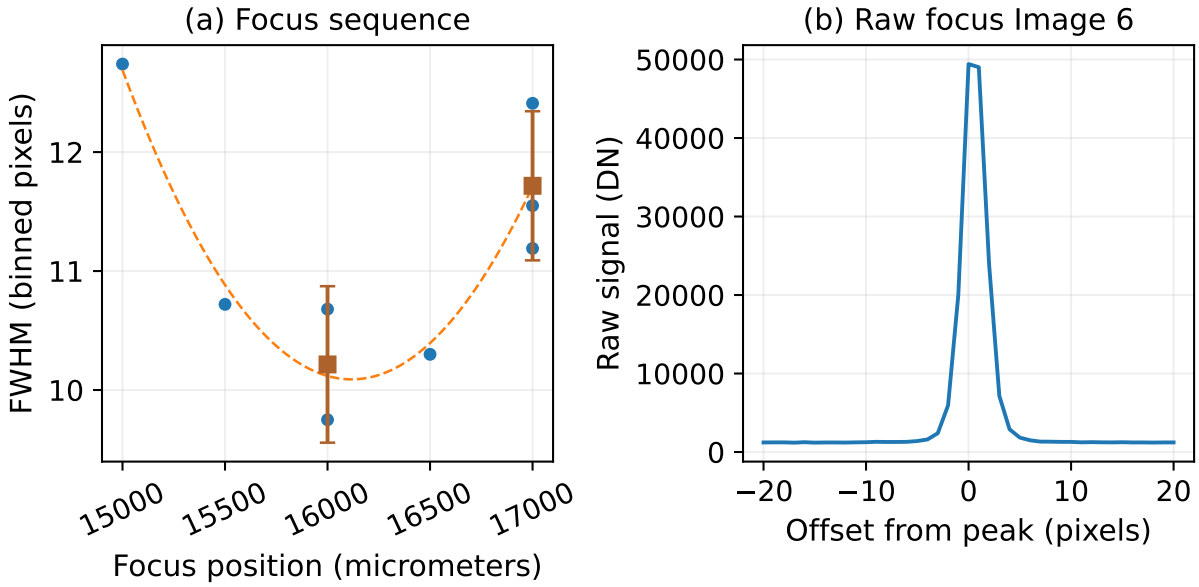


Figure 6: (a) FWHM versus focus, with sample-SD bars at repeated positions and a descriptive quadratic. (b) A row through the brightest pixel in raw focus Image 6; the flattened core limits the measured width.

Table 7: AstroImageJ FWHM measurements of Omicron Aquilae at different focus positions.

CCD Image	Focus (μm)	FWHM (pixels)
1	17000	11.55
2	17000	12.41
3	17000	11.19
4	16500	10.30
6	16000	9.75
7	16000	10.68
8	15500	10.72
9	15000	12.74

0.30%. Both estimates are below the observed 1.33% scatter, indicating that source photon statistics alone cannot account for the variation. Tracking, changes in transparency, flat-field residuals, and correlations introduced during calibration may also contribute.

3.3 Exposure Time, Width, and Count Rate

I measured twelve unbinned images from 6 to 300 s using a 30-pixel aperture and 45–75-pixel sky annulus (Table 8). Image 25 is 150 s according to its header. Figure 10 shows that FWHM increases from 6.79 to 22.97 pixels, while count rate falls from about 2.98×10^5 to 6.99×10^4 DN s^{-1}. Stable unsaturated stellar images should have approximately constant width and count rate. The longer images are elongated (Figure 11) and have flattened cores, consistent with tracking motion and nonlinear response.

Raw stellar-region peaks rise from 27401 DN at 6 s to 42552 DN at 8 s, then remain near 45900–46400 DN through 150 s (47300 DN at 300 s). No raw frame reaches 65535 DN. The roughly 45000-DN calibrated plateau therefore does not demonstrate ADC clipping; analog

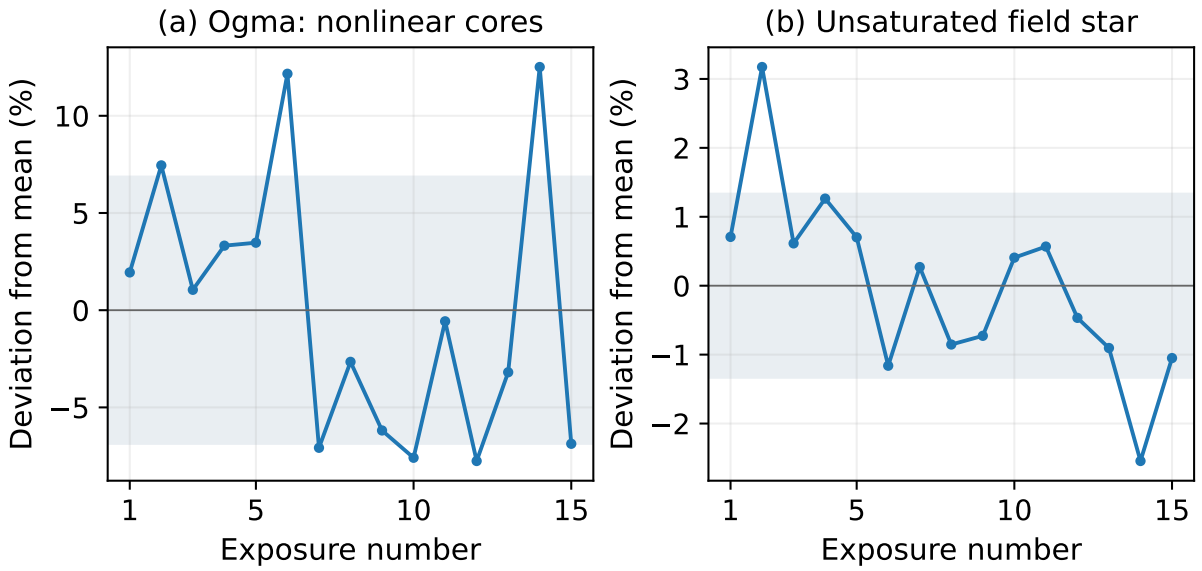


Figure 7: Count-rate deviations from the fifteen-frame means: (a) nonlinear Ogma cores, 6.86% sample scatter; (b) the unsaturated field star, 1.33%. Shading spans one sample SD. These are single-exposure scatters, not errors on the means.

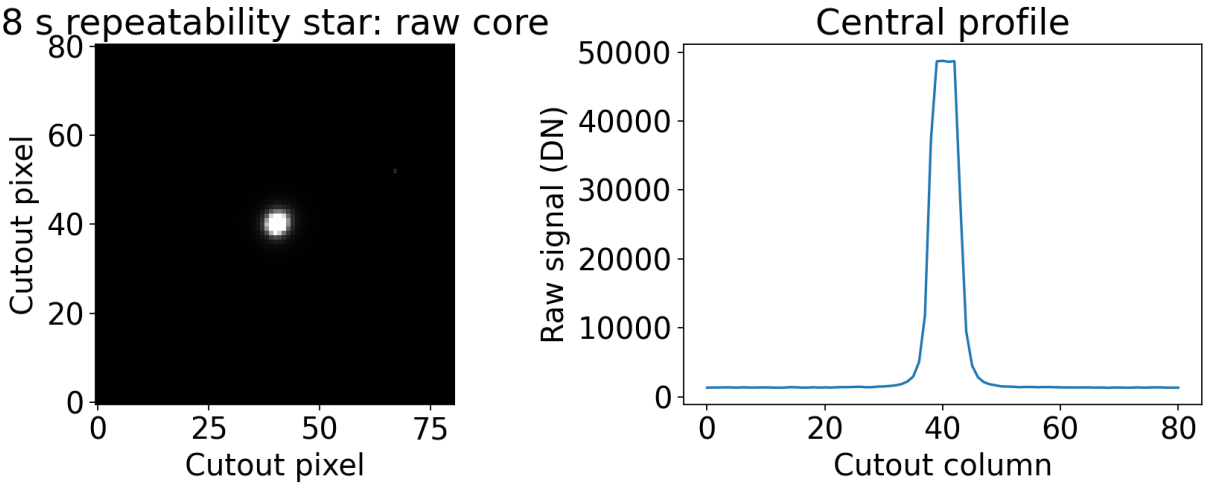


Figure 8: Ogma repeatability image and central profile. The flattened core near 49000 DN limits bright-star photometry.

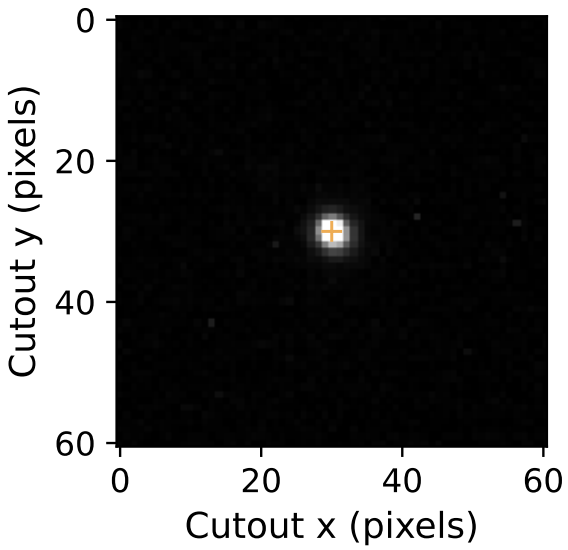


Figure 9: Identification cutout of the supplementary unsaturated field star. Its count-rate deviations are shown in Figure 7(b).

nonlinearity or another acquisition limit remains possible. Different calibration and acquisition settings change the reported peak values, so the focus, repeatability, and dome images do not establish one common saturation threshold.

For Images 21–26, a 30-pixel aperture recovers 3.6–9.7% more flux than a 17-pixel aperture, although their different annuli also affect the comparison. Although the larger aperture recovers additional flux, it does not remove the decline in count rate. This indicates that aperture losses alone cannot explain the observed behavior; nonlinear response, tracking, and possible changes in transparency must also be considered. Individual errors for this exposure-time series were unavailable.

4 Exercise D: Color Photometry

4.1 M29 Measurements

We observed M29 through B, V, and R with fifteen 30-s frames per filter (450 s per band). Separate normalized dome flats were applied in AstroImageJ. The saved R-band setup applies a master bias and R flat with dark subtraction disabled; the master-flat combination method was not recorded. The calibrated images were aligned and averaged within each band.

I measured twelve stars with an 8-pixel aperture and 15–22-pixel sky annulus. Stars 10 and 11 reach about 48981 DN in R and were excluded, leaving ten. The finding-chart labels (Figure 12) match Tables 9 and 10: Star 1 is aperture T1, and the other stars retain their AstroImageJ C-numbers.

For background-subtracted counts C in an average stack, $m_{\mathrm{inst}} = -2.5 \log_{10}(C/t)$. Averaging retains the single-frame signal scale, so $t = 30$ s; the equal exposure factor cancels:

$$(B - V)_{\mathrm{inst}} = -2.5 \log_{10}(C_B/C_V), \tag{22}$$

$$(V - R)_{\mathrm{inst}} = -2.5 \log_{10}(C_V/C_R). \tag{23}$$

I propagated the exported formal count errors using

$$\sigma_m = \frac{2.5}{\ln 10} \frac{\sigma_C}{C}, \tag{24}$$

then added independent band variances for each color. The shared V measurement gives $\mathrm{Cov}(B - V, V - R) = -\sigma_V^2$. These formal errors exclude calibration and atmospheric systematics. Averaging fifteen independent images reduces random noise by $\sqrt{15}$, but shared calibration errors and alignment correlations remain. Be-

Table 8: Ogma exposure-time measurements using a 30-pixel photometric aperture. Peak values are the reported values from the measurement images; they should not be interpreted as a universal raw-detector saturation threshold.

Image	Time (s)	FWHM (pixels)	Rate (DN s^{-1})	Peak (DN)
15	6	6.79	297312	26361
18	8	6.64	299378	41512
17	10	7.30	275574	44927
19	12	7.83	242015	44972
16	15	8.61	237392	44877
20	20	9.22	197273	45015
21	45	12.50	159497	45064
22	60	14.50	158589	44998
23	90	16.36	131787	45139
24	120	17.50	113424	45165
25	150	17.24	87429	45385
26	300	22.97	69921	45897

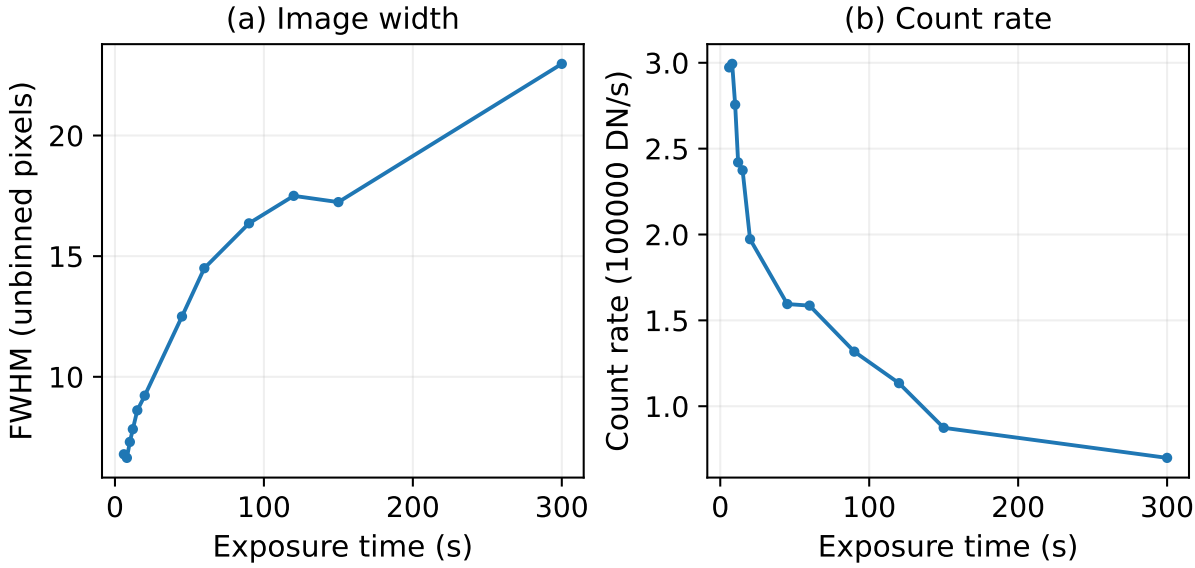


Figure 10: Ogma FWHM and count rate versus exposure time, using unbinned pixels and a 30-pixel aperture. Per-image uncertainties are unavailable. Longer exposures broaden and lose measured count rate.

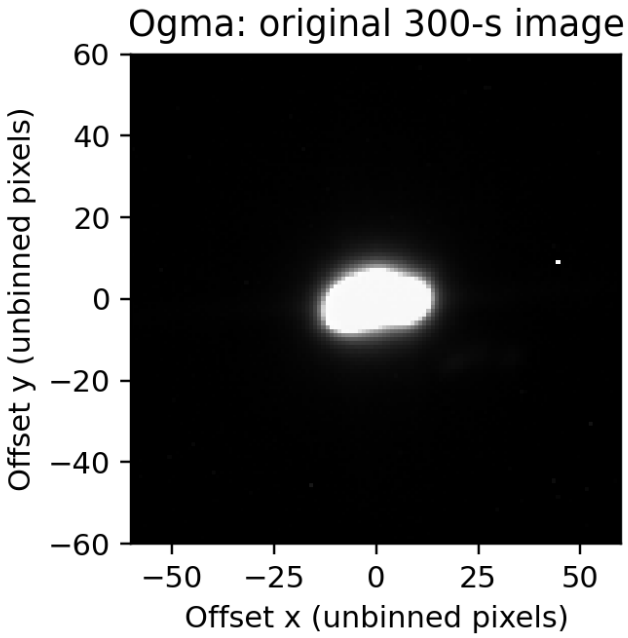


Figure 11: Original 300-s Ogma image cutout, showing elongation.

cause the saved stacks lack a verified variance model, the exported errors are retained as formal estimates and are not used to infer intrinsic color scatter.

Figure 13 shows that stars with larger B–V generally have larger V–R. Most B–V values lie between 0.81 and 1.09 mag; Star 8 is redder at $B-V = 1.529$ and $V-R = 0.572$ mag. It follows the trend and was retained. Temperature, interstellar reddening, and field-star contamination can all affect the colors [5, 6]. Without standard stars, zero points, or color transformations, these instrumental indices do not directly give temperatures or spectral types. Membership data and a verified stack-error model would strengthen the interpretation.

4.2 NGC 7129 Composite

The Python pipeline retained 18 B, 15 V, and 20 R dome flats, excluding B frames 9 and 19 and V frames 4, 6, 8, 16, and 20. Each flat was median-normalized; the median-combined master was renormalized to unity. Science frames were already dark-corrected. I removed their stored 100-DN pedestal by adding the signed `PEDESTAL=-100` value, then divided by the matching flat, without further bias or dark subtraction.

I aligned to the first V frame with `astroalign` and median-combined fifteen frames per filter. To reduce a broad background gradient, I subtracted a model made with a 31×31 median filter and Gaussian smoothing ($\sigma = 120$ pixels). Each channel was median-subtracted, clipped at zero, scaled by its 99.5th percentile, and stretched with an inverse hyperbolic sine. R, V, and B map to red, green, and blue.

Figure 14 shows the nebula's structure and bright central region. Because the channels were scaled independently, the displayed colors are qualitative. In addition, the smooth background model may remove some faint extended nebulosity. These processing choices limit the image to a morphological comparison rather than a quantitative color or surface-brightness measurement. Masking both stars and nebulosity when estimating the background would better preserve the diffuse emission.

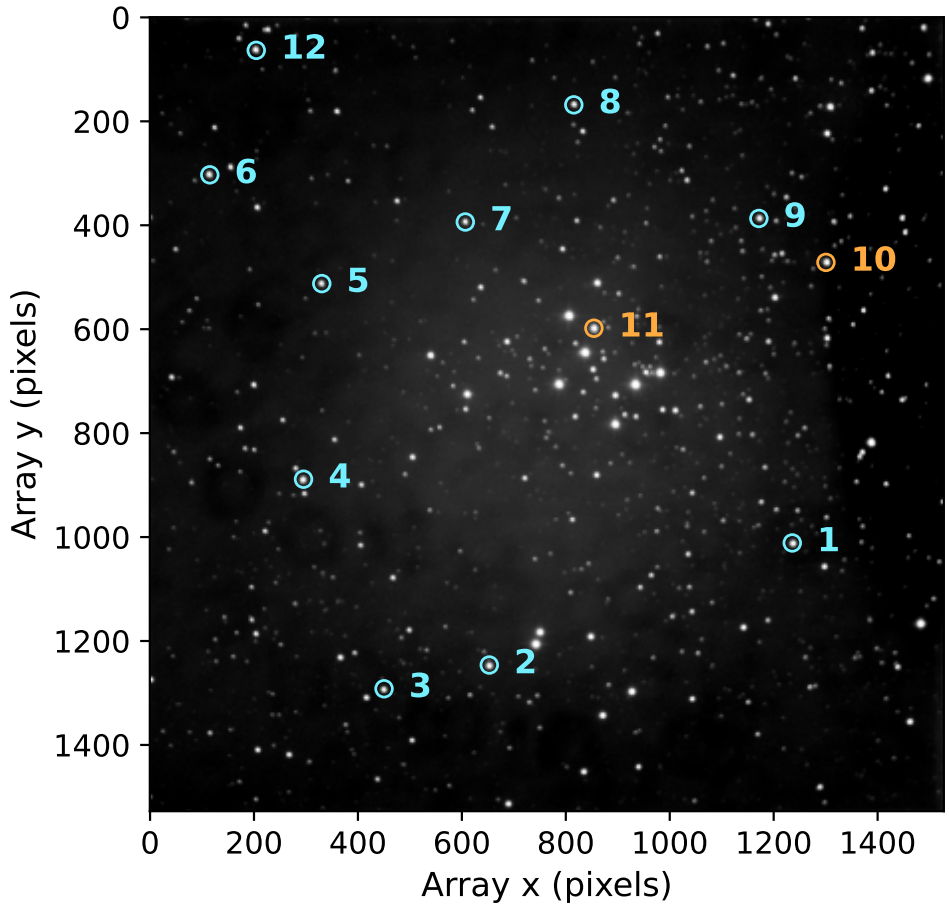


Figure 12: M29 V-stack finding chart with the measured aperture positions. Cyan labels identify the ten retained stars; orange labels mark excluded Stars 10 and 11. Array orientation is shown, without an assumed celestial scale.

Table 9: Background-subtracted M29 signals and formal AstroImageJ uncertainties in DN, rounded to the nearest DN.

Star	B (DN)	V (DN)	R (DN)
1	272429 ± 671	578024 ± 967	692679 ± 1019
2	258869 ± 664	608546 ± 986	811736 ± 1079
3	152103 ± 574	348257 ± 840	455550 ± 896
4	231023 ± 643	521315 ± 942	681979 ± 1019
5	160092 ± 582	367103 ± 853	467630 ± 904
6	205532 ± 619	548129 ± 951	731946 ± 1037
7	119655 ± 550	252116 ± 787	300119 ± 808
8	76431 ± 504	312512 ± 819	529038 ± 934
9	175259 ± 594	478265 ± 915	679195 ± 1012
12	256088 ± 657	602996 ± 978	756523 ± 1046

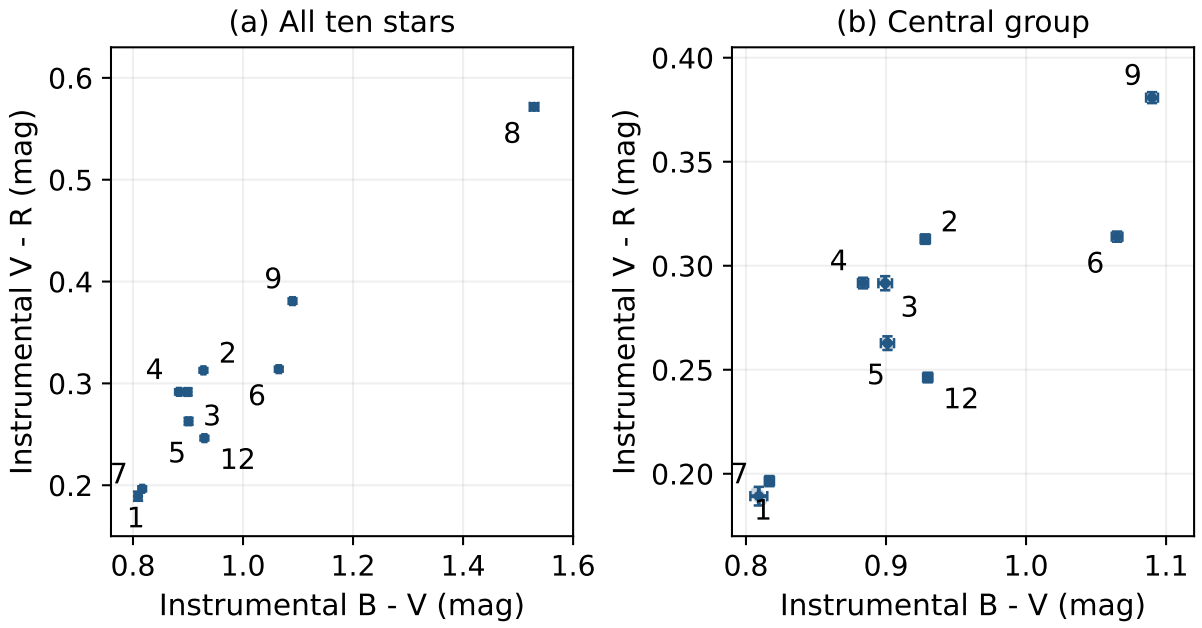


Figure 13: Instrumental color-color diagrams: (a) all ten retained stars; (b) an enlargement of the central group, with Star 8 outside the view. Bars propagate formal exported errors. The faint ellipses in (b) show the formal two-color covariance at an ellipse radius of one standard deviation; their negative tilt comes from the shared V measurement. Identifiers match the finding chart and tables.

Table 10: Instrumental color indices of the 10 selected stars in the M29 field.

Star	$B - V$ (mag)	$V - R$ (mag)
1	0.817 ± 0.003	0.196 ± 0.002
2	0.928 ± 0.003	0.313 ± 0.002
3	0.899 ± 0.005	0.292 ± 0.003
4	0.884 ± 0.004	0.292 ± 0.003
5	0.901 ± 0.005	0.263 ± 0.003
6	1.065 ± 0.004	0.314 ± 0.002
7	0.809 ± 0.006	0.189 ± 0.004
8	1.529 ± 0.008	0.572 ± 0.003
9	1.090 ± 0.004	0.381 ± 0.003
12	0.930 ± 0.003	0.246 ± 0.002

5 Exercise E: Plate Scale

5.1 Observations and Identifications

On September 24 we observed ten double-star targets, mostly through [O III], with 2×2 binning (1528 × 1528 pixels). Most had three 30-s exposures; HR6446 also had 60-s exposures. The observing plan (Figure 15) records the targets, but its old separations are not the WDS values used in the calibration.

I verified HR6200, HR7751, HR8815, and HR7310 against DSS2 red fields [4]. Affine matches to 27, 33, 13, and 22 surrounding stars gave RMS residuals of 1.8, 1.3, 1.4, and 1.2 arcseconds. The companion directions agreed with the catalog. These matches support the component identifications, although the saturated survey primaries limit precise position measurements.

5.2 Separations and Plate Scale

AstroImageJ measured HR6200, HR7751, and HR8815 in three exposures each. HR6200 and HR8815 used 9-pixel primary and 4-pixel companion apertures with 40–60-pixel annuli; HR7751 used 6-pixel apertures and 19–26-pixel annuli. For HR7310, Python used a 12-pixel-radius background-subtracted centroid for the primary and a robust Gaussian plus planar-background fit to the faint companion in a 17×17 box.



Figure 14: BVR composite of NGC 7129 after flat correction, alignment, median stacking, and broad background subtraction. Channel scaling is for display.

ID	Name	Type	RA	Dec	'Azim...	Altitu...	Rise	Transit ▲	Set	Vis	Mag	Ma...	Sep
HR5618	SAO45357, HD1336...	Dbl	15h 03m 47s	+47°39.2'	317°	20°	5:03 AM	4:03 PM	3:04 AM	Yes	4.8	5.6	0.7
HR6200	SAO46210, HD1504...	Dbl	16h 38m 45s	+48°55.6'	308°	34°	Circum	5:38 PM	Circum	Yes	4.9	10.3	25.6
HR6212	SAO65485, HD1506...	Dbl	16h 41m 17s	+31°36.1'	291°	26°	9:28 AM	5:41 PM	1:53 AM	Yes	2.8	5.4	1.1
HR6446	SAO160479, HD156...	Dbl	17h 20m 50s	-12°50.8'	249°	5°	1:03 PM	6:20 PM	11:38 P...	Yes	4.3	9.1	46.3
HR7310	SAO18222, HD1807...	Dbl	19h 12m 33s	+67°39.7'	333°	54°	Circum	8:12 PM	Circum	Yes	3.1	12.3	88.1
HR7751	SAO49385, HD1929...	Dbl	20h 15m 28s	+47°42.8'	298°	69°	10:13 A...	9:15 PM	8:17 AM	Yes	4.0	9.5	208.9
HR7942	SAO70467, HD1979...	Dbl	20h 45m 40s	+30°43.1'	246°	70°	1:38 PM	9:45 PM	5:53 AM	Yes	4.2	9.2	6.1
HR7947	SAO106475, HD197...	Dbl	20h 46m 39s	+16°07.4'	222°	60°	2:45 PM	9:46 PM	4:47 AM	Yes	5.1	6.1	9.8
HR8285	SAO190559, HD206...	Dbl	21h 42m 01s	-23°15.7'	187°	26°	6:05 PM	10:42 PM	3:18 AM	Yes	5.2	11.4	5.1
HR8815	SAO128001, HD218...	Dbl	23h 09m 32s	+08°40.6'	153°	55°	5:36 PM	12:09 AM	6:42 AM	Yes	5.1	10.6	32.9

Figure 15: Original AstroPlanner observing plan for ten targets. The calibration uses the later WDS separations and epochs in Table 11.

For each exposure,

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}, \quad p = \theta/d, \tag{25}$$

where θ is the matching WDS separation [3]. Table 11 gives the repeated measurements. Figure 16 compares pixel and catalog separations. The four pair means range from 1.232 to 1.283 arcsec pixel^{-1}, exceeding the repeat SEMs. Equal weighting gives

$$\boxed{\bar{p} = 1.251 \pm 0.012 \text{ arcsec pixel}^{-1}}, \tag{26}$$

with between-target SD 0.023 and SEM SD/$\sqrt{4}$. The reference measurements span catalog epochs from 2015 to 2021 and have no quoted separation uncertainties in the adopted tables. Changes in binary separation, centroid bias, and field distortion may therefore contribute to the spread. With only four verified pairs, these contributions cannot be separated. HR7751's longer pixel baseline gives the most repeatable result and agrees with the mean.

5.3 Remaining Targets

Figure 17 shows all ten observed fields, but only four have verified calibration measurements. Table 12 lists the status of all ten targets. HR6446's candidate gives separations of 34.335 and 35.774 pixels; using the 45.6-arcsecond planning value would give diagnostic scales of 1.328 and 1.275. Its component identification and centering are not established, and a third frame has an artifact, so it is excluded. Only the four verified pairs enter the adopted plate scale.

For HR7310 I use 80.4 arcseconds (WDS 2021), rather than the 88.1-arcsecond planning value. The original AstroImageJ angular distances encode a fixed 0.1109116 arcsec pixel^{-1} setting; I instead used pixel coordinates and catalog separations.

5.4 Image Width and Field of View

Six unsaturated companion measurements in HR6200 and HR8815 (peaks 1273–1702 DN) have mean FWHM 3.471 pixels and sample SD 0.318. Multiplying by the plate scale gives an observed width of $\boxed{w = 4.34 \text{ arcsec}}$, with sample scatter $\boxed{s_w = 0.40 \text{ arcsec}}$; the scale SEM contributes 0.04 arcsecond to the mean width. Because focus, tracking, and pixel sampling also broaden stellar images, this value describes the observed angular image width rather than an isolated measurement of atmospheric seeing.

The side length of the nominal full-array field is

$$L = 1528\bar{p}/60 \tag{27}$$

where L is in arcminutes. The field spans $\boxed{31.87 \times 31.87 \text{ arcmin}}$; each side has a 0.29-arcminute contribution from the scale SEM. The side errors are fully correlated. This excludes distortion and cropped borders.

Table 11: Verified binary measurements. Uncertainties in mean pixel distance and mean plate scale are SEMs of three exposures; they exclude catalog and shared systematic errors.

Target	WDS separation (arcsec; year)	Individual distances (pixels)	Mean distance (pixels)	Mean scale (arcsec pixel^{-1})
HR6200	28.1; 2020	22.557, 22.791, 22.746	22.698 ± 0.072	1.2380 ± 0.0039
HR7751	208.5; 2015	166.458, 166.340, 166.492	166.430 ± 0.046	1.2528 ± 0.0003
HR8815	32.6; 2015	25.422, 25.490, 25.306	25.406 ± 0.054	1.2832 ± 0.0027
HR7310	80.4; 2021	65.050, 65.406, 65.404	65.287 ± 0.118	1.2315 ± 0.0022

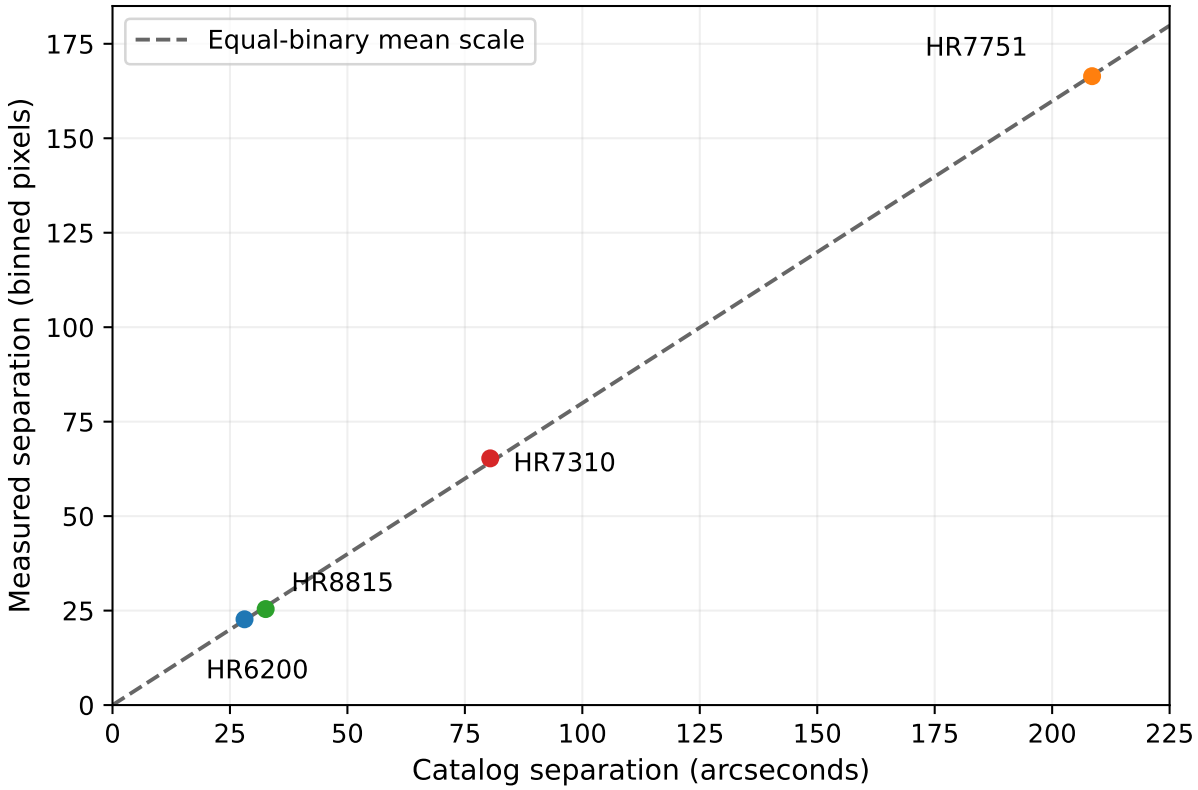


Figure 16: Mean pixel separation versus catalog angular separation. Repeat SEMs are smaller than the markers. The dashed line uses the equal-binary mean scale, not a weighted fit. Catalog errors are unavailable.

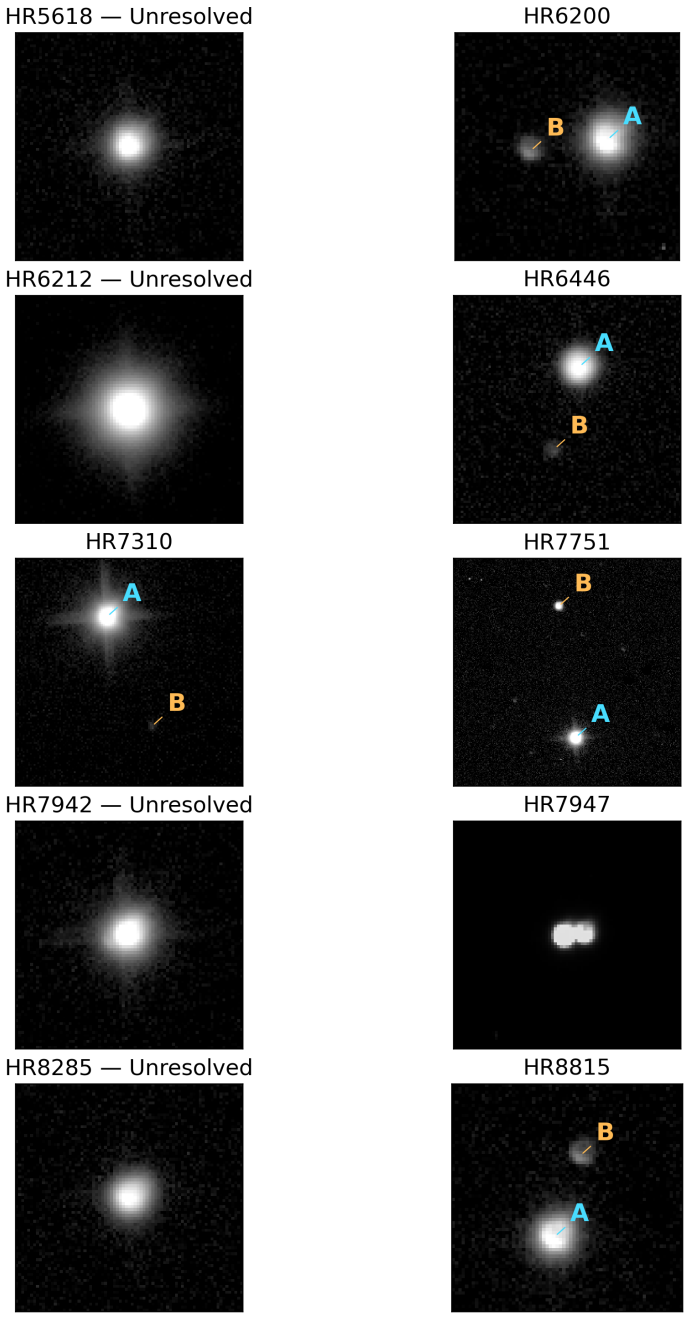


Figure 17: The ten observed fields. Markers show initial selections; only the four verified pairs enter the calibration. HR7310 uses the corrected companion. Panels have different stretches and crops and do not share a calibrated celestial orientation.

Table 12: Status of all ten observed double-star targets. Dashes indicate that no verified calibration value is available.

Target	Separation (pixels)	Used?	Status
HR5618	—	No	Unresolved in these images.
HR6200	22.698	Yes	Verified repeated measurements; Table 11.
HR6212	—	No	Companion identification uncertain.
HR6446	34.335, 35.774	No	Candidate only; inconsistent centering and one artifact frame.
HR7310	65.287	Yes	Corrected companion identification.
HR7751	166.430	Yes	Verified repeated measurements.
HR7942	—	No	Companion identification uncertain.
HR7947	—	No	Overlapping cores; reliable separate centroids unavailable.
HR8285	—	No	Companion identification uncertain.
HR8815	25.406	Yes	Verified repeated measurements.

Conclusion

I have characterized the CCD response and examined how calibration and observing conditions affect stellar measurements. The paired bias frames give a read noise of 10.045 ± 0.007 DN, while the joint flat-pair fit gives a binned gain of 1.290 ± 0.006 $e^- DN^{-1}$. Dark signal increases with exposure time and temperature, consistent with a thermal contribution. However, the unmatched bias and uncertain detector temperatures limit the interpretation of the inferred dark current and activation energy.

The stellar observations locate best focus near 16000 μm, although nonlinear cores limit the measured minimum width. An unsaturated field star shows 1.33% repeatability scatter, exceeding the source photon-noise prediction. The broadening and declining count rate in longer Ogma exposures further demonstrate the combined effects of nonlinear response, tracking, and aperture losses. The M29 instrumental colors identify Star 8 as relatively red, while the NGC 7129 composite reveals nebular structure without providing calibrated colors. Finally, four verified binaries give a plate scale of 1.251 ± 0.012 arcseconds per pixel and a 31.87×31.87-arcminute field. More complete temperature records, unsaturated stellar exposures, and additional verified binary measurements would provide stronger constraints on these results.

References

[1] ASTRO 320W, *Observing Exercise Manual*, version August 22, 2026.

[2] Diffraction Limited, *SBIG STX and STXL Large Format CCD Detectors*, technical specifications, December 2019. https://cdn.diffractionlimited.com/downloads/STX-STXLOverview.pdf

[3] U.S. Naval Observatory, *Washington Double Star Catalog*, summary tables for 12–17 h and 18–23 h, accessed October 20, 2026. https://www.astro.gsu.edu/wds/

[4] Space Telescope Science Institute / CDS, *Digitized Sky Survey 2, red survey*, accessed through the CDS HiPS2FITS service, October 20, 2026. https://alasky.cds.unistra.fr/hips-image-services/hips2fits

[5] OpenStax, *Astronomy 2e*, Section 17.2, "Colors of Stars." https://openstax.org/books/astronomy-2e/pages/17-2-colors-of-stars

[6] OpenStax, *Astronomy 2e*, Section 20.3, "Cosmic Dust." https://openstax.org/books/astronomy-2e/pages/20-3-cosmic-dust

Appendix A: Analysis Code and Supporting Files

The accompanying code package contains the analysis notebooks, measurement exports, figures, and report source. The repository is `https://github.com/gracemarie07/ASTRO320W_Midterm_Davis`. The submitted package's README lists the required inputs and execution instructions. Raw observing files are stored separately. All notebook names below refer to the `notebooks/` directory unless a `scripts/` path is given.

Table 13: Guide to the main analysis files. Notebook names refer to the `notebooks/` directory.

Exercise	Main files and purpose
A	`A320_bf_stdev_final.ipynb` and `A320_bf_stdev_ROI.ipynb`: instructor read-noise algorithms. `A_dark_signal.ipynb`: dark trends.
B	`B_gain_workbook.ipynb`, `B_photon_transfer_curves.ipynb`: paired gain and PTCs. The original 10-s workbook is in `spreadsheets/`.
C	`exercise_C_omicron_aquilae.ipynb`, `exercise_C_ogma.ipynb`, `C_unsaturated_field_star.ipynb`: focus, bright-star photometry, and supplementary repeatability.
D	`D_M29_colors.ipynb`, `D_NGC7129.ipynb`: instrumental colors and the BVR composite.
E	`E_field_identifications.ipynb`, `E_supplementary_centroids.ipynb`, `E_plate_scale.ipynb`, `E_observed_fields.ipynb`: identification, centroids, scale, and image panels.
Supplement	`Report_revision.ipynb` runs `scripts/revise_analysis.py`: reconciled M29 measurements, joint gain fit, noise maps, raw peak checks, and revised plate-scale plot. `scripts/flat_profiles.py` produces the identified flat comparisons.

The two `A320_bf_stdev` algorithms were supplied by the instructor. The observations were collected with the Artemis group; I performed the analysis and wrote the report. ChatGPT assisted with code organization and revision, consistency checks between the data and report, supplementary calculations and plots, and editing the report and documentation.